

Numerical Selection of Forecast-Optimal Smoothness in Penalized Trend Estimation

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Abstract

Selecting a smoothing parameter for penalized trend estimation becomes a numerical problem when the criterion is chronological forecast loss rather than in-sample fit. We reparameterize finite-difference penalized least squares by a normalized smoothness coordinate $S \in [0, 1]$ and minimize a rolling forecast-validation objective that may contain multiple local minima. The proposed derivative-aware search combines adaptive subdivision, endpoint refinement, Brent root refinement, and exact boundary evaluation. With the algorithm frozen before the final runs, it recovered 240/240 relevant adversarial optima, 2105/2105 synthetic reference minima, and 473/473 minima on financial-series stress tests, while using about 1.6–1.8% of the dense-grid evaluations in the latter two benchmarks. Applied case studies show that multimodality is configuration-dependent rather than universal and that the validation ranking of frozen minima can change on a later test block.

Keywords: penalized least squares; smoothing parameter; trend estimation; rolling-origin evaluation; numerical optimization; time series.

1 Introduction

Quadratic penalized smoothers provide a simple way to estimate a slowly varying component from noisy observations. Their basic form goes back to Whittaker’s graduation method [12] and appears in time-series trend estimation as penalized least squares (PLS) [7]. The smoothing parameter λ trades fidelity to the observed series against smoothness of the estimated trend. Small values allow the trend to follow the observations closely, whereas large values increasingly suppress penalized finite differences.

The choice of λ has therefore received substantial attention. Classical approaches include cross-validation (CV) and generalized cross-validation (GCV) [5]; Cortés-Toto et al. [4] implement CV, GCV, AICc, and BIC directly in the PLS framework and study the smoothness attained by each criterion. The controlled-smoothness line instead parameterizes the effect of λ through a trace-based smoothness index that can be interpreted in percentage terms [4, 8]. Efficient numerical implementations have also been developed for Whittaker–Henderson smoothing and GCV [11], while Franke et al. [6] calibrate the Hodrick–Prescott smoothing parameter against known simulated trends.

Predictive selection of trend smoothness is also established. Hart [9] introduced time-series cross-validation for kernel trend smoothing, selecting the bandwidth from one-step-ahead prediction errors based on past observations. Thus, the contribution here is not the idea of choosing smoothness by predictive performance. More generally, forecast evaluation has a well-developed temporal vocabulary: forecast origins can be fixed or rolling, models can be recalibrated as the origin advances, and training windows can be expanding or rolling [1, 10].

Nor is the possibility of multiple minima unique to the present criterion. Biessy [2] shows a Whittaker–Henderson example in which a GCV profile has two local minima while studying marginal-likelihood selection and numerical optimization. The problem considered here starts from these established ingredients and

asks a narrower numerical question: for finite-difference PLS, how can the multiple relevant minima of a horizon-specific rolling future-block forecast objective be recovered efficiently and compared with the exact limiting smoothing models?

The difficulty is numerical as well as statistical. The resulting forecast-validation objective is one-dimensional, but it need not have a single local minimum. A coarse grid may miss relevant minima, while a very dense grid is reliable only at a substantial evaluation cost. Searching directly in $\lambda \in [0, \infty)$ also requires arbitrary finite bounds and devotes numerical resolution unevenly across smoothing regimes.

This paper studies that numerical problem. The PLS estimator is reparameterized by a normalized smoothness coordinate $S \in [0, 1]$, derived from the controlled-smoothness index but rescaled so that the two limiting smoothing regimes are represented exactly by $S = 0$ and $S = 1$. Analytic first and second derivatives of the rolling forecast loss are combined with a derivative-aware adaptive search that detects and classifies relevant stationary points, refines derivative roots with Brent's method [3], and compares all detected local minima with the exact smoothness-domain endpoints.

The contribution is deliberately narrow. We do not propose a new smoother or a new forecasting model, and we do not optimize jointly over model order, window length, and smoothness. Instead, we ask whether forecast-optimal smoothness can be located accurately and with substantially fewer evaluations than a dense reference search when the forecast-loss surface may contain multiple local minima. Recovering those alternatives is practically relevant because separated minima can represent different trend scales and forecast paths even when their validation errors are relatively close. The method is validated on analytic adversarial objectives, 1920 synthetic rolling forecast objectives, and 384 real financial-data objective surfaces. A separate development/test case study then examines how the recovered minima translate into fitted trends and future forecasts without using the test block for selection.

2 Penalized trend estimation and forecast validation

2.1 Penalized least squares

Let $y \in \mathbb{R}^N$ denote an observed time series and let D_d be the matrix representing the d -th finite-difference operator, with $1 \leq d < N$. Define

$$Q = D_d^\top D_d.$$

For a smoothing parameter $\lambda \geq 0$, the PLS trend estimate is

$$\hat{\tau}_\lambda = \arg \min_{\tau \in \mathbb{R}^N} \{ \|y - \tau\|_2^2 + \lambda \|D_d \tau\|_2^2 \} = H_\lambda y, \quad (1)$$

where

$$H_\lambda = (I + \lambda Q)^{-1} \quad (2)$$

is the smoothing matrix. This is the finite-difference PLS formulation used by Guerrero [7] and belongs to the broader Whittaker–Henderson family [11, 12].

The matrix Q has nullity d . Consequently, the limit $\lambda \rightarrow \infty$ does not eliminate the null-space component of the trend. Rather,

$$H_\lambda \longrightarrow P_{\ker(D_d)},$$

the orthogonal projector onto the null space of the difference penalty. For $d = 1$ this is the constant subspace, for $d = 2$ it is the discrete linear subspace, and higher orders give the corresponding polynomial-like limits.

2.2 A normalized smoothness coordinate

The controlled-smoothness literature uses the trace of the smoothing matrix to quantify attained smoothness [4, 8]. For $d \geq 1$, let

$$S_G(\lambda; N) = 1 - \frac{\text{tr}(H_\lambda)}{N}.$$

Because Q has d zero eigenvalues, S_G ranges from 0 to $1 - d/N$. For numerical search we normalize this quantity to the full unit interval:

$$S(\lambda) = \frac{N}{N-d} S_G(\lambda; N) = 1 - \frac{1}{N-d} \sum_{j=1}^{N-d} \frac{1}{1 + \lambda \delta_j}, \quad (3)$$

where $\delta_1, \dots, \delta_{N-d} > 0$ are the positive eigenvalues of Q . Thus

$$S(0) = 0, \quad \lim_{\lambda \rightarrow \infty} S(\lambda) = 1.$$

The coordinate is strictly increasing because

$$S'(\lambda) = \frac{1}{N-d} \sum_{j=1}^{N-d} \frac{\delta_j}{(1 + \lambda \delta_j)^2} > 0, \quad (4)$$

and it is concave:

$$S''(\lambda) = -\frac{2}{N-d} \sum_{j=1}^{N-d} \frac{\delta_j^2}{(1 + \lambda \delta_j)^3} < 0. \quad (5)$$

Equation (3) therefore replaces the unbounded penalty domain $[0, \infty]$ by the compact smoothness domain $[0, 1]$ without changing the ordering of smoothing levels.

2.3 Rolling forecast-validation objective

Smoothing the historical sample and extrapolating the fitted trend are separate operations. In this paper the continuation rule is fixed to the native finite-difference continuation associated with the penalty order. Let $G_{d,h}$ denote the linear operator that maps a fitted trend of length L to its next h continued values by imposing zero d -th finite difference beyond the forecast origin.

At forecast origin T , let

$$x_T = (y_{T-L+1}, \dots, y_T)^\top$$

be the rolling training window and

$$z_T = (y_{T+1}, \dots, y_{T+h})^\top$$

the future validation block. The trend forecast is

$$\hat{z}_T(\lambda) = G_{d,h} H_\lambda x_T,$$

where H_λ now has dimension $L \times L$. With residual

$$r_T(\lambda) = z_T - \hat{z}_T(\lambda),$$

the origin-specific loss is

$$f_T(\lambda) = \frac{1}{h} \|r_T(\lambda)\|_2^2.$$

For a set $\mathcal{O}$ of rolling forecast origins, the pooled forecast-validation objective is

$$f_{d,L,h}(\lambda) = \frac{1}{|\mathcal{O}|} \sum_{T \in \mathcal{O}} f_T(\lambda). \quad (6)$$

The trend estimator is refit at every origin using only the L observations available at that origin. The selection problem is

$$S^* \in \arg \min_{S \in [0,1]} F_{d,L,h}(S), \quad F_{d,L,h}(S(\lambda)) = f_{d,L,h}(\lambda). \quad (7)$$

Thus, forecast-optimal in this paper always means optimal with respect to the rolling future-block mean squared error in (6).

2.4 Why several forecast-CV minima can matter

Let

$$Q = U \operatorname{diag}(\delta_j) U^\top.$$

In this basis the smoother attenuates penalized components through

$$\alpha_j(\lambda) = \frac{1}{1 + \lambda \delta_j}. \quad (8)$$

The attenuation is component-specific: as λ changes, different penalized directions disappear at different rates. Consequently, the fitted endpoint level, slope, and higher-order finite differences can change substantially across smoothing levels. The continuation operator then maps those endpoint features into future values.

This provides a mechanism by which several local minima can arise. A rougher trend can preserve recent movement and extrapolate a stronger local slope or curvature, while a smoother trend can suppress that movement and extrapolate a more stable low-frequency path. Their forecast errors can cross across rolling origins, so the average objective in (6) need not be unimodal. The existence of multiple minima is therefore a property to diagnose rather than an assumption imposed on every series.

2.5 Analytic derivatives

The derivative of the smoothing matrix follows from differentiating a matrix inverse:

$$H'_\lambda = -H_\lambda Q H_\lambda, \quad H''_\lambda = 2H_\lambda Q H_\lambda Q H_\lambda.$$

Define

$$a_T(\lambda) = G_{d,h} H_\lambda Q H_\lambda x_T,$$

and

$$b_T(\lambda) = G_{d,h} H_\lambda Q H_\lambda Q H_\lambda x_T.$$

Then

$$f'_T(\lambda) = \frac{2}{h} r_T(\lambda)^\top a_T(\lambda), \quad (9)$$

$$f''_T(\lambda) = \frac{2}{h} [\|a_T(\lambda)\|_2^2 - 2r_T(\lambda)^\top b_T(\lambda)]. \quad (10)$$

The derivatives of the pooled objective are obtained by averaging (9)–(10) over forecast origins.

The monotone change of coordinate preserves the interior stationary points:

$$F'(S) = \frac{f'(\lambda)}{S'(\lambda)}. \quad (11)$$

Moreover,

$$F''(S) = \frac{f''(\lambda)}{[S'(\lambda)]^2} - \frac{f'(\lambda)S''(\lambda)}{[S'(\lambda)]^3}. \quad (12)$$

At a stationary point $f'(\lambda^*) = 0$, the second term vanishes and the sign of the curvature is preserved. We can therefore search in the compact S -domain while evaluating analytic derivatives through λ .

3 Adaptive search on the smoothness domain

The numerical target is the set of relevant local minima of $F(S)$, together with the exact smoothness-domain endpoints. The algorithm does not assume that F is unimodal.

The frozen procedure begins with nine uniformly spaced points over the interior search interval and augments the first and last coarse cells with six levels of deterministic dyadic refinement. The additional endpoint

resolution is important because the map $S(\lambda)$ compactifies an unbounded λ -domain; distinct features at large λ can therefore become compressed close to $S = 1$.

Each sampled point provides the objective, gradient, and curvature in the smoothness coordinate through (11)–(12). An interval is subdivided when the sampled derivatives indicate unresolved structure: a turning gradient, a change in curvature sign, a midpoint derivative that is small relative to the endpoint derivatives, or a pattern consistent with more than one sign change inside the interval. Subdivision stops at maximum depth eight or interval width 10^{-3} . The frozen derivative and curvature tolerances are 10^{-8} , the relative near-zero threshold is 0.2, and the interior upper boundary is $1 - 10^{-6}$.

After adaptive sampling, every derivative sign change is bracketed and refined with Brent’s method [3] to a root tolerance of 10^{-10} . Stationary points are classified by the sign of $F''(S)$. The global selected value is the best among all detected local minima and the two exact endpoint candidates:

$$S = 0 \iff \lambda = 0, \quad S = 1 \iff \lambda = \infty.$$

The $S = 1$ objective is evaluated through the exact null-space limit rather than by replacing one with a value such as 0.999999.

When several nearby minima are useful for downstream inspection, a post-processing radius ε may be applied after discovery. Candidates are ordered by objective value and worse minima within ε of an accepted minimum are suppressed. Because this operation occurs after root discovery and ranking, it does not change the selected best minimum. The numerical benchmarks below therefore assess the search before this optional summarization step.

No finite adaptive sampler can certify recovery of every stationary point of an arbitrary smooth scalar function. The empirical validation is consequently designed around missed local minima, selected-smoothness error, positive objective regret, and evaluation cost rather than a universal root-recovery claim.

4 Experimental design

4.1 Adversarial analytic objectives

Nine deterministic polynomial objectives were constructed directly in the S -domain with known stationary structure. They include minima near both endpoints, two close minima, two close upper-tail minima, a flat quartic minimum, five separated minima, exact boundary optima, and a stationary inflection that is not a local minimum. Each shape was transformed through the $S \leftrightarrow \lambda$ map for

$$N \in \{63, 126, 252, 504\}, \quad d \in \{1, 2, 3, 4\},$$

giving 144 method configurations. The frozen adaptive search was compared with a 20 001-point dense S -grid and a 257-point uniform search in $\log \lambda$, with derivative roots refined after bracketing.

4.2 Synthetic forecast-validation surfaces

Synthetic observations follow a local-linear latent trend with AR(1) observation noise. The latent slope evolves as a random walk and the observed series is the latent trend plus stationary AR(1) noise. Three regimes were used: baseline ($\sigma_{\Delta\text{slope}} = 0.01, \sigma_\eta = 0.5, \phi = 0.3$), persistent noise (0.01, 0.5, 0.8), and a rougher latent trend (0.03, 0.5, 0.3).

Each series has 800 observations. The paper-scale benchmark crosses ten random seeds with

$$d \in \{1, 2, 3, 4\}, \quad L \in \{63, 126, 252, 504\}, \quad h \in \{1, 5, 20, 60\},$$

for a total of 1920 forecast-validation surfaces. Rolling windows advance in steps of five observations and the last 30 valid origins are scored. Each adaptive result is compared with a 5001-point dense reference grid on $[0, 1]$.

4.3 Real-data geometry stress test

The final external check uses six versioned daily closing-price series: SPY, QQQ, AAPL, XOM, BTC–USD, and ETH–USD. The final 2000 positive observations of each tracked snapshot are transformed to natural logarithms. The same four orders, four window lengths, and four horizons used in the synthetic benchmark produce

$$6 \times 4 \times 4 \times 4 = 384$$

real-data objective surfaces. These experiments test numerical geometry only; they are not a comparison of financial forecasting models and are not interpreted as evidence of return or price predictability.

4.4 Applied development/test case studies

To interpret the numerical minima, a separate applied experiment uses four pre-specified tracked series: quarterly U.S. real GDP (GDPC1), SPY, AAPL, and BTC–USD. The final eight quarters of GDP and the final 60 observations of each daily market series are reserved before any configuration search. All positive levels are modeled on the natural-log scale.

Within the development region, the pre-specified (d, L, h) grid is scanned using the frozen search. Candidate minima are summarized with the already frozen spacing radius $\varepsilon = 0.10$, retaining at most three representatives ranked by rolling forecast-CV error. Among configurations with at least two representatives, the displayed case maximizes the smoothness separation between its two lowest-CV candidates, with lexicographic (d, L, h) tie breaking. This rule uses development data only.

After the candidate set is frozen, each candidate is refit at the final development origin and scored on the first h observations of the untouched test block. A validation–test rank reversal is recorded when the lowest-CV candidate is not the lowest-test-error candidate among this already frozen set. The test ranking is retrospective diagnostic information; it is never fed back into candidate or configuration selection.

4.5 Frozen protocol and sensitivity

The primary search specification was frozen after a pre-paper quick benchmark and before the 1920-surface confirmatory run. A separate one-factor-at-a-time sensitivity analysis then varied the initial grid size, maximum depth, endpoint-refinement levels, minimum interval width, and near-zero derivative threshold while leaving all other settings fixed.

A dense-grid local minimum is counted as recovered when an adaptive local minimum lies within two dense-grid cells. Positive objective regret is

$$\max\{F(\hat{S}) - F(S_{\text{ref}}), 0\},$$

with values below 10^{-12} treated as numerical roundoff. Because the adaptive root refinement is continuous while the reference is discrete, a negative raw regret can occur when the adaptive method locates a lower point between two grid nodes.

5 Results

Table 1 summarizes the three frozen benchmark families. The adaptive search recovered every relevant known adversarial minimum or boundary optimum (240/240), every dense-reference interior minimum in the synthetic benchmark (2105/2105), and every dense-reference interior minimum in the financial stress test (473/473). No benchmark case had positive objective regret beyond the numerical threshold.

Table 1: Frozen benchmark summary. Evaluation fraction is the mean adaptive evaluation count divided by the corresponding dense-grid count.

Benchmark	Surfaces	Minima recovered	Positive-regret cases	Mean evals.	Dense evals.	Eval. fraction	Max $ \Delta S^* $
Adversarial	144	240/240	0	75.2	20001	0.38%	9.06×10^{-5}
Synthetic	1920	2105/2105	0	78.4	5001	1.57%	1.02×10^{-4}
Financial	384	473/473	0	91.9	5001	1.84%	9.98×10^{-5}

On the synthetic forecast surfaces, the adaptive search required 78.35 evaluations on average versus 5001 for the dense reference, or 1.57% of the dense evaluation count. The maximum difference between the selected adaptive smoothness and the dense-grid choice was 1.02×10^{-4} , approximately one half of a dense-grid cell. Total measured runtime was 27.29 seconds for the adaptive search and 1497.85 seconds for the dense reference on the benchmark machine. The wall-clock ratio is implementation- and hardware-dependent, so the evaluation-count comparison is the more portable computational result.

The financial stress test showed similar behavior: 91.89 adaptive evaluations on average versus 5001 dense evaluations, an evaluation fraction of 1.84%, with maximum selected- S discrepancy 9.98×10^{-5} . Measured total runtime was 6.52 seconds for the adaptive search and 292.97 seconds for the dense reference.

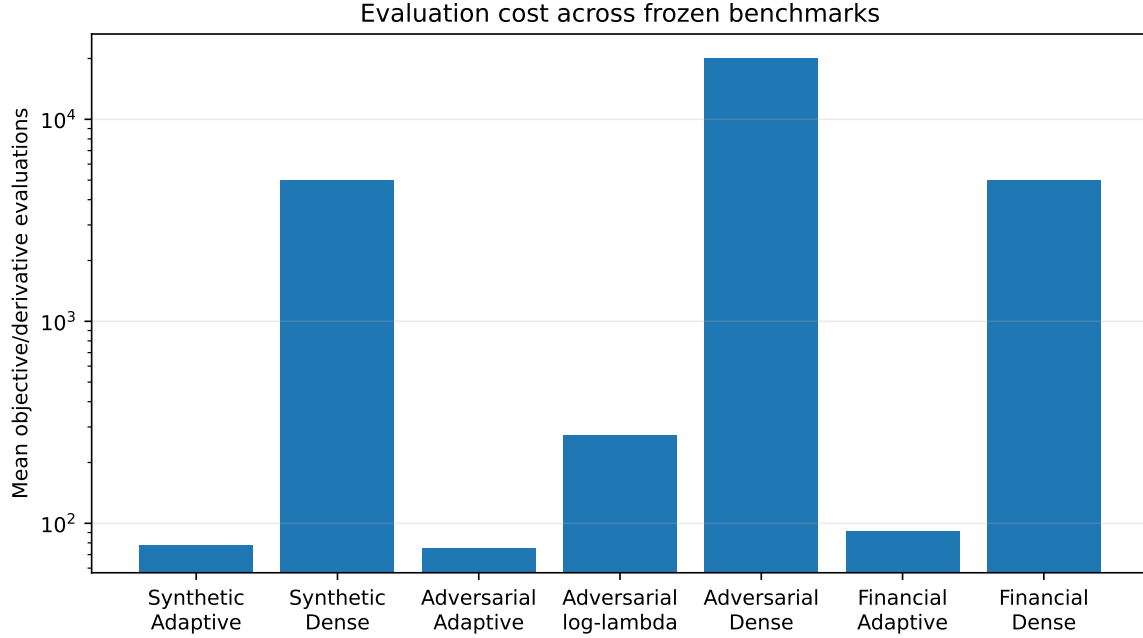


Figure 1: Mean evaluation counts across the frozen benchmarks. The vertical axis is logarithmic. One adaptive evaluation returns the objective and its first two derivatives, whereas a dense-grid evaluation requires only the objective; measured wall-clock times are therefore reported separately in the text.

Figure 2 compares the smoothness selected by the adaptive search with the dense reference for all synthetic and financial surfaces. The points are concentrated on the identity line throughout the unit interval, including cases where the optimum lies near a smoothness-domain endpoint.

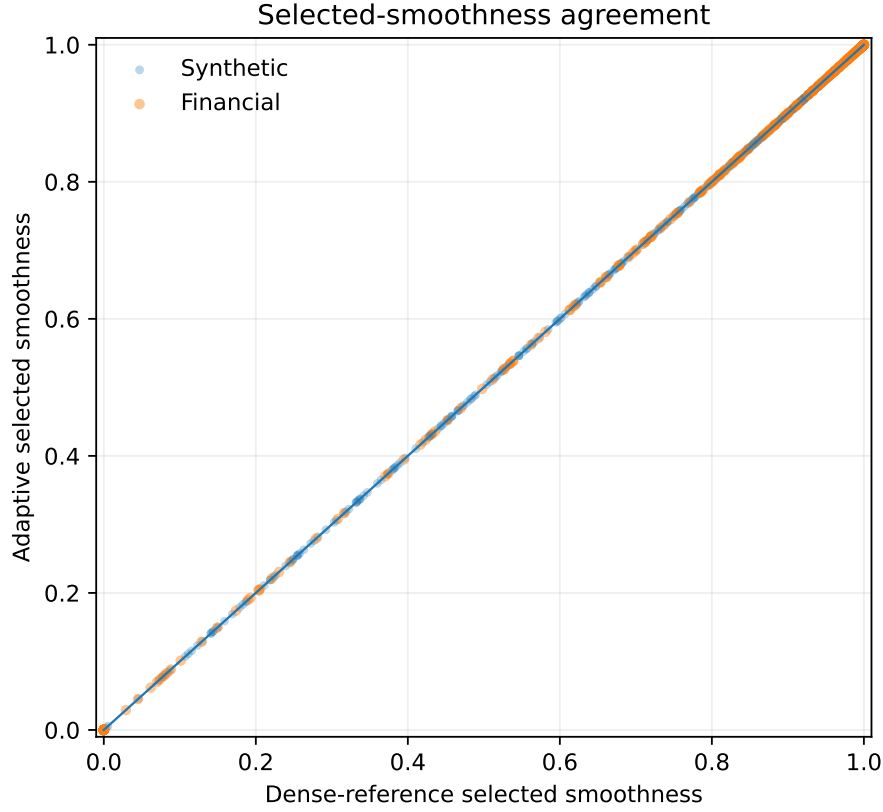


Figure 2: Agreement between adaptive and dense-reference selected smoothness for the 1920 synthetic and 384 financial objective surfaces.

Exact endpoint handling was not merely formal. In the synthetic benchmark, the selected optimum was interior in 1574 surfaces, $S = 0$ in 312, and $S = 1$ in 34. In the financial stress test, the corresponding counts were 321, 53, and 10. Replacing $S = 1$ by a large but finite λ would therefore alter a non-negligible part of the benchmark protocol.

The adversarial benchmark also separates the proposed search from a simple alternative parametrization. The uniform $\log \lambda$ stationary search detected 226 of 240 relevant known minima or boundary optima and required 273.44 evaluations on average. Its misses were concentrated in the deliberately close-minimum constructions. The adaptive S -search detected all 240 with 75.24 mean evaluations. The stationary-inflection target was not detected by the adaptive method or by the dense local-minimum detector, but it is not an optimization candidate and is outside the primary claim.

The sensitivity study explains which parts of the frozen specification matter. Table 2 reports the one-factor variants. Initial grid sizes from 5 to 13, maximum depths from 4 to 10, tested minimum interval widths, and endpoint refinement of at least two levels all recovered every reference minimum in the 216-surface sensitivity panel. In contrast, removing endpoint refinement missed five minima, produced positive regret in three cases, and increased the maximum selected- S error to 1.85×10^{-2} . Reducing the near-zero derivative ratio from 0.2 to 0.1 missed one secondary local minimum, although it did not change the selected global optimum.

Table 2: One-factor-at-a-time sensitivity around the frozen primary specification. The sensitivity study is diagnostic; it was not used to retune the confirmatory algorithm after the final benchmark.

Specification	Missed minima	Positive-regret cases	Mean evals.	Max $ \Delta S^* $
Primary	0	0	82.8	9.92×10^{-5}
initial_grid_size=5	0	0	75.4	9.92×10^{-5}
initial_grid_size=7	0	0	83.0	9.92×10^{-5}
initial_grid_size=13	0	0	93.8	9.92×10^{-5}
max_depth=4	0	0	71.5	9.92×10^{-5}
max_depth=6	0	0	80.0	9.92×10^{-5}
max_depth=10	0	0	82.8	9.92×10^{-5}
endpoint_refinement_levels=0	5	3	64.0	1.85×10^{-2}
endpoint_refinement_levels=2	0	0	70.2	9.92×10^{-5}
endpoint_refinement_levels=4	0	0	75.5	9.92×10^{-5}
endpoint_refinement_levels=8	0	0	90.8	9.92×10^{-5}
min_interval=0.0005	0	0	88.8	9.92×10^{-5}
min_interval=0.002	0	0	76.8	9.92×10^{-5}
min_interval=0.005	0	0	70.8	9.92×10^{-5}
near_zero_ratio=0.1	1	0	82.1	9.92×10^{-5}
near_zero_ratio=0.5	0	0	92.7	9.92×10^{-5}

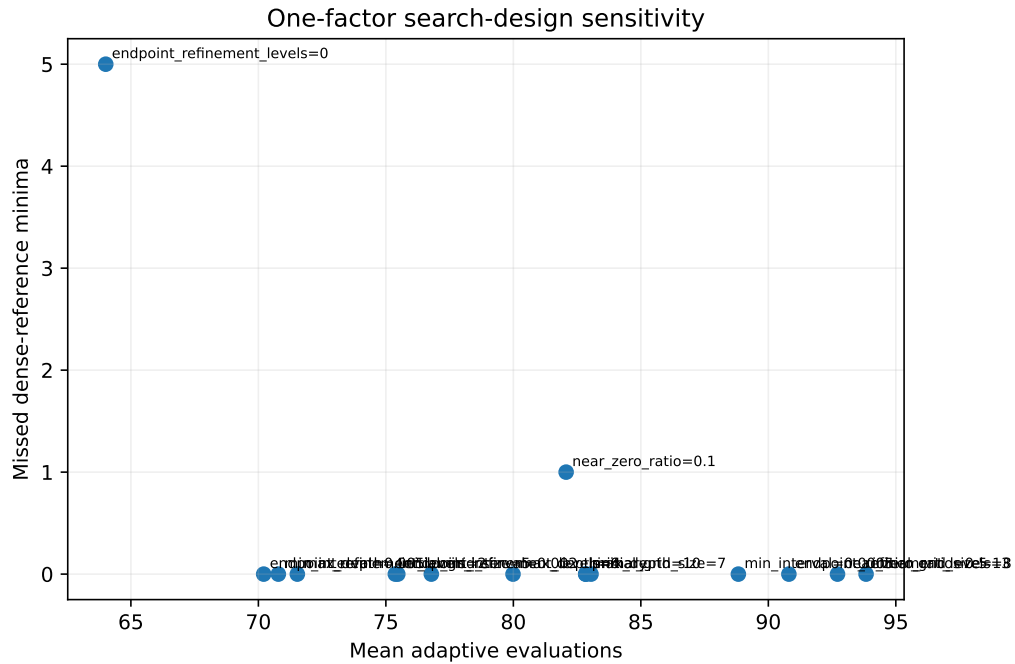


Figure 3: Evaluation cost and missed-reference-minimum count in the one-factor sensitivity study. The main observed failure mode is the removal of endpoint-aware refinement.

The optional ε -spacing rule affects only how multiple detected minima are summarized. With $\varepsilon = 0.10$, 359 nearby candidates were suppressed across 330 synthetic surfaces, while the best detected minimum in each surface was preserved by construction.

5.1 Applied interpretation of multiple minima

The applied scan confirms that multimodality is not universal. None of the 64 pre-specified GDP configurations produced two ε -separated representative minima. In contrast, this occurred in 7/64 SPY configurations,

10/64 AAPL configurations, and 9/64 BTC–USD configurations; three BTC configurations produced three representatives. GDP therefore provides a useful unimodal contrast to the market examples.

The selected SPY case used $d = 1$, $L = 126$, and $h = 60$. Rolling CV ranked an interior minimum at $S = 0.95639$ first, with MSE 0.00177070, and the exact lower endpoint $S = 0$ second, with MSE 0.00195813. Thus the endpoint candidate had about 10.6% larger validation error. After freezing these two candidates, their test MSE values were 0.00093985 and 0.00066164, respectively. The validation ordering therefore reversed: the second-ranked validation candidate had about 29.6% lower test MSE on the later block.

The AAPL case produced two widely separated minima, $S = 0.62901$ and $S = 0.99733$, but the second had much larger validation and test error. The selected BTC–USD case produced three representatives at $S = 0.84463$, 0.02908, and 0.99786. Its second candidate was only about 13.9% worse than the winner in rolling CV, yet its test MSE was about 14.7 times larger. These examples show that separated minima can range from competitive alternatives to clearly inferior local optima.

The test comparison is not used to replace rolling CV by retrospective test selection. Its role is to reveal selection sensitivity: when several local minima exist, the validation winner need not dominate every later realization. Recovering the candidate set therefore provides information that a single unimodal optimization run would discard.

6 Discussion

The numerical benefit of the normalized smoothness coordinate is not that it makes the forecast objective convex. It does not. Its value is that it turns an unbounded penalty parameter into a compact, interpretable domain whose endpoints correspond to exact limiting estimators. Because $S'(\lambda) > 0$, stationary points and their local minimum/maximum classification are preserved in the interior. This allows the search to operate in a meaningful smoothness coordinate while derivative calculations continue to exploit the simple matrix dependence on λ .

Compactification also creates a numerical issue: a large range of λ is compressed near $S = 1$. The early diagnostic runs showed that several derivative roots can occupy the same coarse upper-tail cell. Endpoint-aware refinement was introduced in response to that observed geometry, then frozen before the paper-scale benchmark. The subsequent sensitivity experiment independently reproduces the failure when endpoint refinement is removed. This is a useful example of a transformation that improves global search organization while still requiring explicit attention to its boundary geometry.

The paper’s scope is narrower than general smoothing-parameter selection. CV, GCV, information criteria, marginal likelihood, controlled smoothness, and simulation-calibrated HP parameters remain legitimate answers to their respective statistical objectives [2, 4–6, 8]. Here the target is specifically rolling future-block forecast MSE. The contribution is therefore the numerical treatment of that criterion when multiple local minima are possible, not a claim that forecast loss is universally preferable to the alternatives. The applied cases also show why retaining several detected minima can be informative: they form a finite set of distinct smoothing choices whose validation ranking may or may not persist out of sample. This candidate set should be interpreted as a selection-sensitivity diagnostic, not as a confidence set or probability distribution over smoothness.

Several limitations remain. First, no finite sampler can guarantee detection of every stationary root of an arbitrary smooth function without additional assumptions. The adversarial stationary inflection illustrates this distinction: it is stationary but not a local minimum. Second, the dense reference grid is a benchmark rather than mathematical ground truth. Third, the analytic derivative formulas rely on a fixed affine trend-continuation operator. If additional forecasting parameters were re-estimated as functions of S , for example inside an ARIMA continuation model, the derivative expressions would change. Finally, the financial experiment is only a real-data numerical stress test and provides no evidence of trading value or financial predictability.

The implementation, frozen experiment scripts, tracked data snapshots, and result summaries are available in the project repository at <https://github.com/heri-espino/Trend-Estimation>. The final benchmark inputs are versioned so the figures and tables in this paper can be regenerated from fixed result directories.



Figure 4: Applied interpretation of forecast-CV minima. Each row corresponds to GDP, SPY, AAPL, or BTC-USD. The left panels show the development-region forecast-CV profile and representative minima; the center panels show trend estimates fitted to the same final development window; the right panels compare the frozen candidate forecasts with the untouched test block. Candidate labels retain validation rank even when the later test ranking differs.

7 Conclusions

Forecast-optimal smoothness selection for penalized trend estimation can require more than minimizing a unimodal scalar function. By normalizing the controlled-smoothness coordinate to $S \in [0, 1]$, differentiating the chronological forecast-validation objective analytically, and searching adaptively for multiple local minima while retaining exact boundary candidates, the proposed procedure avoids an exhaustive dense scan of the smoothing domain.

Under a protocol frozen before the confirmatory experiments, the method recovered all relevant adversarial optima, all 2105 dense-reference minima across 1920 synthetic forecast surfaces, and all 473 dense-reference minima across 384 financial real-data surfaces. The synthetic and financial benchmarks required on average only 1.57% and 1.84% of the corresponding dense-grid evaluation counts. Sensitivity analysis identified endpoint refinement as the principal structural safeguard near the compactified upper boundary. In the applied case studies, separated minima occurred for several SPY, AAPL, and BTC–USD configurations but not for GDP, and one frozen SPY candidate set exhibited a validation–test rank reversal.

These results support the proposed search as an accurate and computationally economical numerical procedure for the tested rolling forecast-validation problems. They do not establish universal recovery of all stationary points, and the method is intended as a numerical component for forecast-based PLS smoothness selection rather than as a replacement for other statistical smoothing criteria.

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